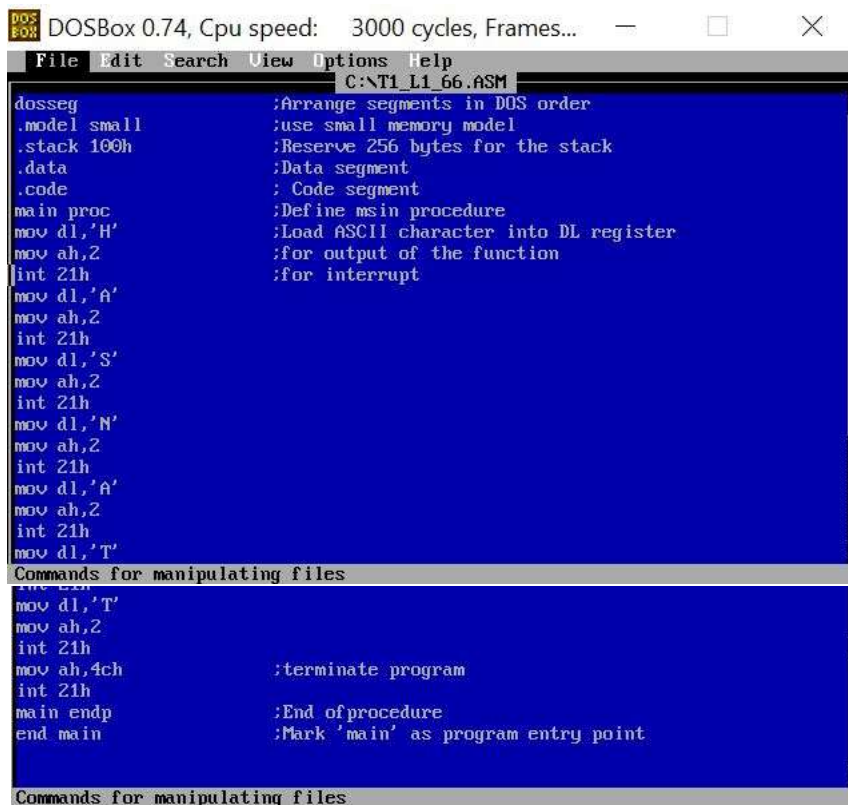
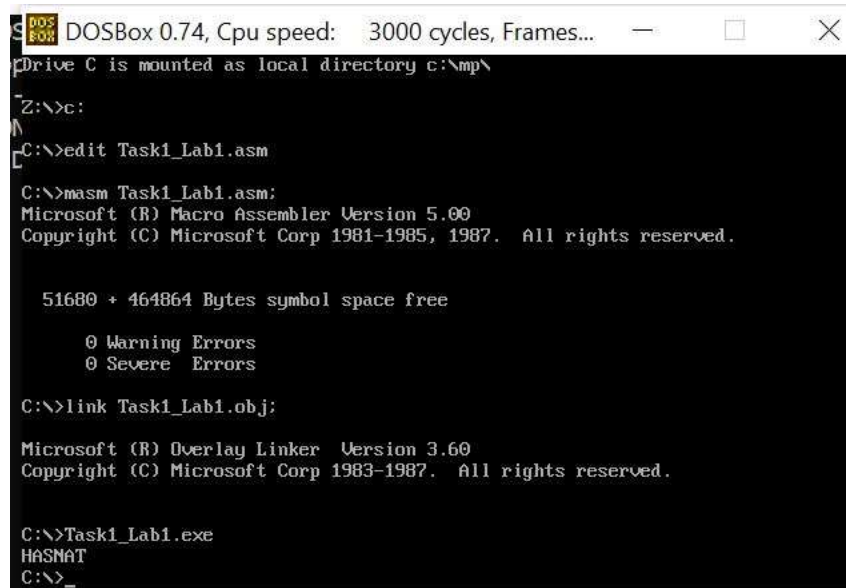


Task 1



```
DOSBox 0.74, Cpu speed: 3000 cycles, Frames...
File Edit Search View Options Help
C:\NT1_L1_66.ASM
dosseg ;Arrange segments in DOS order
.model small ;use small memory model
.stack 100h ;Reserve 256 bytes for the stack
.data ;Data segment
.code ; Code segment
main proc ;Define main procedure
mov dl,'H' ;Load ASCII character into DL register
mov ah,2 ;for output of the function
int 21h ;for interrupt
mov dl,'A'
mov ah,2
int 21h
mov dl,'S'
mov ah,2
int 21h
mov dl,'N'
mov ah,2
int 21h
mov dl,'A'
mov ah,2
int 21h
mov dl,'T'
mov ah,2
int 21h
mov ah,4ch ;terminate program
int 21h
main endp ;End of procedure
end main ;Mark 'main' as program entry point
Commands for manipulating files
```



```
DOSBox 0.74, Cpu speed: 3000 cycles, Frames...
Drive C is mounted as local directory c:\mp\
Z:\>c:
C:\>edit Task1_Lab1.asm
C:\>masm Task1_Lab1.asm:
Microsoft (R) Macro Assembler Version 5.00
Copyright (C) Microsoft Corp 1981-1985, 1987. All rights reserved.

51680 + 464864 Bytes symbol space free

0 Warning Errors
0 Severe Errors

C:\>link Task1_Lab1.obj:
Microsoft (R) Overlay Linker Version 3.60
Copyright (C) Microsoft Corp 1983-1987. All rights reserved.

C:\>Task1_Lab1.exe
HASMAT
C:\>_
```

Task 2



```
DOSBox 0.74, Cpu speed: 3000 cycles, Frames...
File Edit Search View Options Help
C:\TASK2_LAB1.ASM
dosseg                ;Arrange segments
.model small          ;small memory
.stack 100h           ;reserve 256 bytes
.data                 ;data segment
.code                 ;code segment
main proc             ;main procedure
mov dl,'a'            ;Load 'a' in DL register
mov ah,2              ;display character
int 21h               ;interrupt function
mov ah,4ch            ;terminate program
int 21h
main endp             ;end of procedure
end main              ;mark 'main' as program entry points

Commands for manipulating files
```

```
C:\>masm TASK2_LAB1.asm;
Microsoft (R) Macro Assembler Version 5.00
Copyright (C) Microsoft Corp 1981-1985, 1987. All rights reserved.

51680 + 464864 Bytes symbol space free

0 Warning Errors
0 Severe Errors

C:\>link TASK2_LAB1.obj;

Microsoft (R) Overlay Linker Version 3.60
Copyright (C) Microsoft Corp 1983-1987. All rights reserved.

C:\>TASK2_LAB1.EXE
A
C:\>
```

Task 3



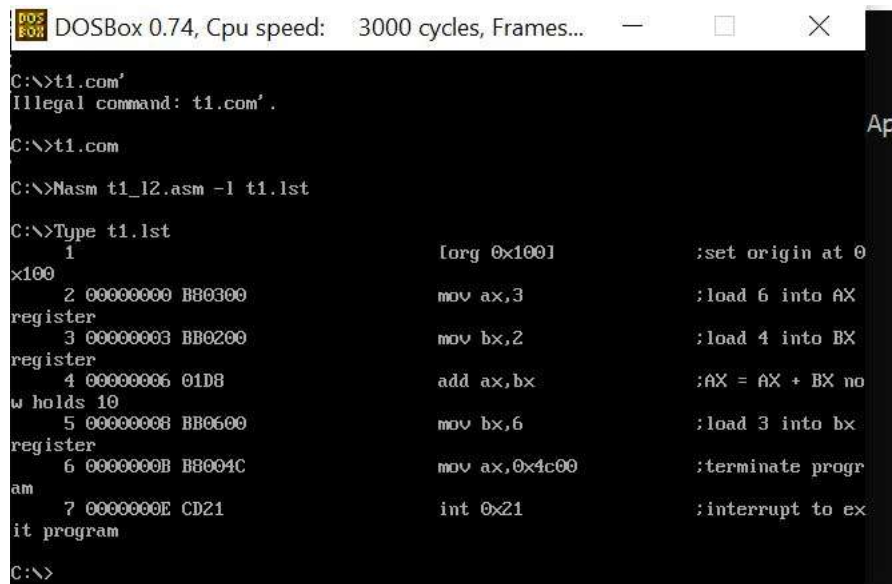
DOSBox 0.74, Cpu speed: 3000 cycles, Frames...

File Edit Search View Options Help

C:\NT1_L2.ASM

```
[org 0x100]      ;set origin at 0x100
mov ax,3         ;load 6 into AX register
mov bx,2         ;load 4 into BX register
add ax,bx        ;AX = AX + BX now holds 10
mov bx,6         ;load 3 into bx register
mov ax,0x4c00    ;terminate program
int 0x21         ;interrupt to exit program
```

Commands for manipulating files



DOSBox 0.74, Cpu speed: 3000 cycles, Frames...

C:\>t1.com'
Illegal command: t1.com'.

C:\>t1.com

C:\>Nasm t1_l2.asm -l t1.lst

C:\>Type t1.lst

```
1 [org 0x100] ;set origin at 0
x100
2 00000000 BB0300 mov ax,3 ;load 6 into AX
register
3 00000003 BB0200 mov bx,2 ;load 4 into BX
register
4 00000006 01D8 add ax,bx ;AX = AX + BX no
w holds 10
5 00000008 BB0600 mov bx,6 ;load 3 into bx
register
6 0000000B BB004C mov ax,0x4c00 ;terminate progr
am
7 0000000E CD21 int 0x21 ;interrupt to ex
it program
```

C:\>

DOSBox 0.74, Cpu speed: 3000 cycles, Frames...

AX 0000 SI 0000 CS 19F5 IP 0100 Stack +0 0000 Flags 7202
BX 0000 DI 0000 DS 19F5 +2 20CD
CX 0010 BP 0000 ES 19F5 HS 19F5 +4 9FFF OF DF IF SF ZF AF PF CF
DX 0000 SP FFFE SS 19F5 FS 19F5 +6 EA00 0 0 1 0 0 0 0 0

CMD >

0100 BB0300	MOV	AX,0003
0103 BB0200	MOV	BX,0002
0106 01D8	ADD	AX,BX
0108 BB0600	MOV	BX,0006
010B BB004C	MOV	AX,4C00
010E CD21	INT	21
0110 89DA	MOV	DX,BX
0112 EB04	JMP	0118

DS:0000	CD 20 FF 9F 00 EA F0 FE	AD DE 1B 05 C5 06 00 00	= f.0= i.+....
DS:0008	AD DE 1B 05 C5 06 00 00	01 01 01 00 02 FF FF FF	f.
DS:0010	18 01 10 01 18 01 92 01	01 01 01 00 02 FF FF FF	f.
DS:0018	01 01 01 00 02 FF FF FF	FF FF FF FF EB 19 C0 11	J.
DS:0020	FF FF FF FF FF FF FF FF	FF FF FF FF 00 00 00 00	J.
DS:0028	FF FF FF FF EB 19 C0 11	00 00 00 00 00 00 00 00	J.
DS:0030	A2 01 14 00 18 00 F5 19	00 00 00 00 00 00 00 00	J.
DS:0038	FF FF FF FF 00 00 00 00	00 00 00 00 00 00 00 00	J.
DS:0040	05 00 00 00 00 00 00 00	00 00 00 00 00 00 00 00	J.
DS:0048	00 00 00 00 00 00 00 00		

1 Step 2ProcStep 3Retrieve 4Help ON 5BRK Menu 6 7 up 8 dn 9 le 10 ri

DOSBox 0.74, Cpu speed: 3000 cycles, Frames...

AX 0003 SI 0000 CS 19F5 IP 0103 Stack +0 0000 Flags 7200
BX 0000 DI 0000 DS 19F5 +2 20CD
CX 0010 BP 0000 ES 19F5 HS 19F5 +4 9FFF OF DF IF SF ZF AF PF CF
DX 0000 SP FFFE SS 19F5 FS 19F5 +6 EA00 0 0 1 0 0 0 0 0

CMD >

0100 BB0300	MOV	AX,0003
0103 BB0200	MOV	BX,0002
0106 01D8	ADD	AX,BX
0108 BB0600	MOV	BX,0006
010B BB004C	MOV	AX,4C00
010E CD21	INT	21
0110 89DA	MOV	DX,BX
0112 EB04	JMP	0118
0114 31D2	XOR	DX,DX

DS:0000	CD 20 FF 9F 00 EA F0 FE	AD DE 1B 05 C5 06 00 00	= f.0= i.+....
DS:0008	AD DE 1B 05 C5 06 00 00	01 01 01 00 02 FF FF FF	f.
DS:0010	18 01 10 01 18 01 92 01	01 01 01 00 02 FF FF FF	f.
DS:0018	01 01 01 00 02 FF FF FF	FF FF FF FF EB 19 C0 11	J.
DS:0020	FF FF FF FF FF FF FF FF	FF FF FF FF 00 00 00 00	J.
DS:0028	FF FF FF FF EB 19 C0 11	00 00 00 00 00 00 00 00	J.
DS:0030	A2 01 14 00 18 00 F5 19	00 00 00 00 00 00 00 00	J.
DS:0038	FF FF FF FF 00 00 00 00	00 00 00 00 00 00 00 00	J.
DS:0040	05 00 00 00 00 00 00 00	00 00 00 00 00 00 00 00	J.
DS:0048	00 00 00 00 00 00 00 00		

1 Step 2ProcStep 3Retrieve 4Help ON 5BRK Menu 6 7 up 8 dn 9 le 10 ri

DOSBox 0.74, Cpu speed: 3000 cycles, Frames...

AX 0003

SI 0000

CS 19F5

IP 0106

Stack +0 0000

Flags 7200

BX 0002

DI 0000

DS 19F5

+2 20CD

CX 0010

BP 0000

ES 19F5

HS 19F5

+4 9FFF

OF DF IF SF ZF AF PF CF

DX 0000

SP FFFE

SS 19F5

FS 19F5

+6 EA00

0 0 1 0 0 0 0 0

CMD >

0103 BB0200

MOV

BX,0002

0106 01DB

ADD

AX,BX

0108 BB0600

MOV

BX,0006

010B BB004C

MOV

AX,4C00

010E CD21

INT

21

0110 89DA

MOV

DX,BX

0112 EB04

JMP

0118

0114 31D2

XOR

DX,DX

0116 31C0

XOR

AX,AX

1

0 1 2 3 4 5 6 7

DS:0000

CD 20 FF 9F 00 EA F0 FE

DS:0008

AD DE 1B 05 C5 06 00 00

DS:0010

18 01 10 01 18 01 92 01

DS:0018

01 01 01 00 02 FF FF FF

DS:0020

FF FF FF FF FF FF FF FF

DS:0028

FF FF FF FF EB 19 C0 11

DS:0030

A2 01 14 00 18 00 F5 19

DS:0038

FF FF FF FF 00 00 00 00

DS:0040

05 00 00 00 00 00 00 00

DS:0048

00 00 00 00 00 00 00 00

2

0 1 2 3 4 5 6 7 8 9 A B C D E F

DS:0000

CD 20 FF 9F 00 EA F0 FE AD DE 1B 05 C5 06 00 00

DS:0010

18 01 10 01 18 01 92 01 01 01 01 00 02 FF FF FF

DS:0020

FF FF FF FF FF FF FF FF FF FF FF FF EB 19 C0 11

DS:0030

A2 01 14 00 18 00 F5 19 FF FF FF FF 00 00 00 00

DS:0040

05 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00

= f.0=

0 1 2 3 4 5 6 7

.....ff.

.....

6.....J.

.....

1 Step

2ProcStep

3Retrieve

4Help ON

5BRK Menu

6

7 up

8 dn

9 le

10 ri